

Statistical Code Review Report

Course: Software Engineering Project 2

Student Name: Dinal Maha Vidanelage, Sandip Ranjit, Swostika Lama, Twe He Gam Aung

Group 9

Date: 15/04/2026

1. Introduction

This report presents the results of a static code analysis performed on a software project using SonarQube and Maven-based SonarScanner integration.

The objective of this analysis is to identify code quality issues, measure software metrics, and evaluate maintainability, reliability, and test coverage.

2. Tools Used

- SonarQube (Code quality dashboard)
- SonarScanner (Maven plugin execution)
- Maven (Build and test execution)
- Java project (analyzed source code)

3. Analysis Method

The analysis was performed using the following command 'mvn clean verify sonar:sonar'.

The results were automatically uploaded to SonarQube dashboard for evaluation of:

- Code quality metrics
- Test coverage
- Code duplication
- Maintainability indicators

4. Summary Metrics

Metric	Value
Test Coverage	
Code Duplication	2.0%
Bugs	0
Vulnerabilities	0
Code Smells	0
Security Rating	A
Reliability Rating	A
Maintainability Rating	A

6. Key Findings

6.1 Test Coverage

The test coverage is 52.7%, which is below the recommended threshold of 80%. This indicates that a significant portion of the codebase is not covered by automated tests.

6.2 Code Quality

The analysis shows no bugs, vulnerabilities, or code smells detected in the project. This indicates that the code is clean, stable, and follows good programming practices.

6.3 Duplication

The duplication rate is 2.0%, which is low and acceptable. This suggests that there is minimal redundant code and the codebase is reasonably well structured.

6.4 Maintainability

The maintainability rating is A, meaning the code has low technical debt and is easy to maintain and extend.

7. Recommendations

- Add missing test cases for uncovered methods
- Maintain current coding standards to preserve code quality
- Continuously run static analysis during development

8: Evidence (Screenshots & Charts)

Figure 1. Screenshot of Overall analysis

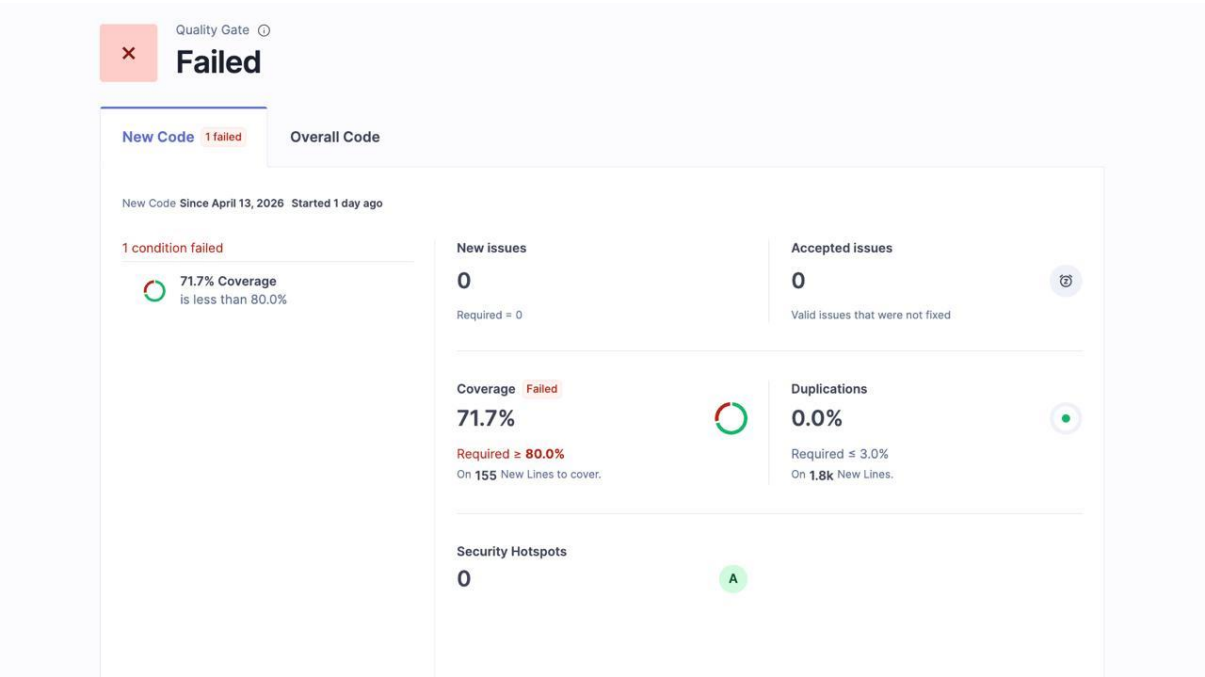


Figure 2. Screenshot of Issues

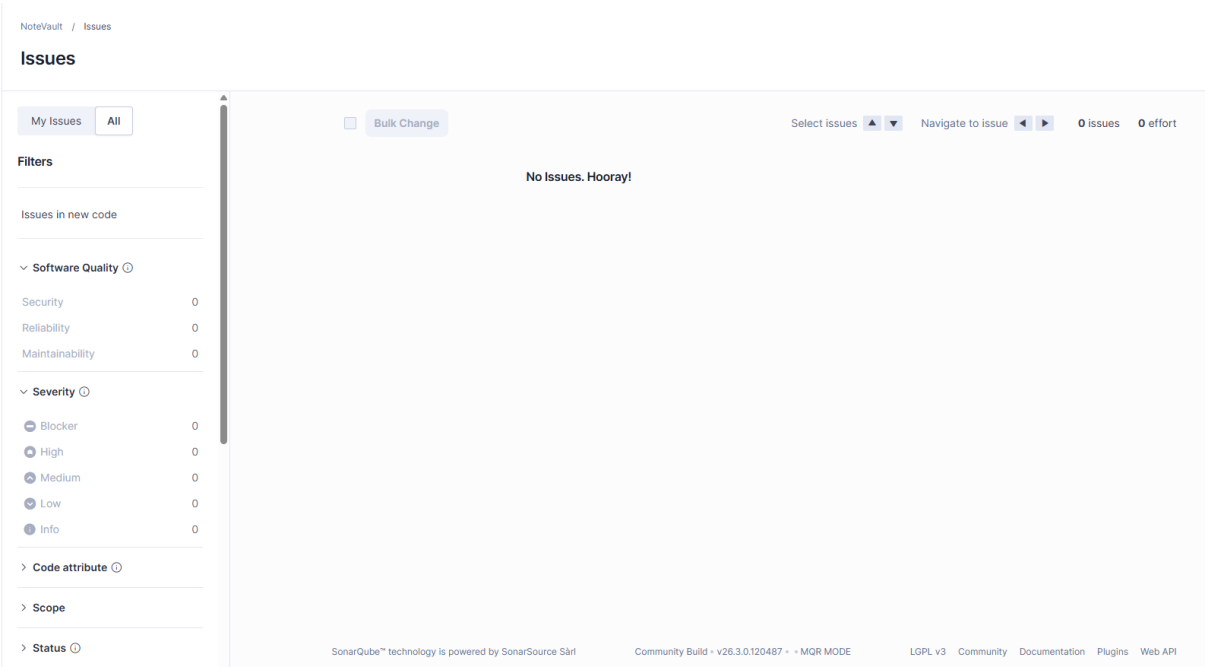


Figure 3. Screenshot of Measures

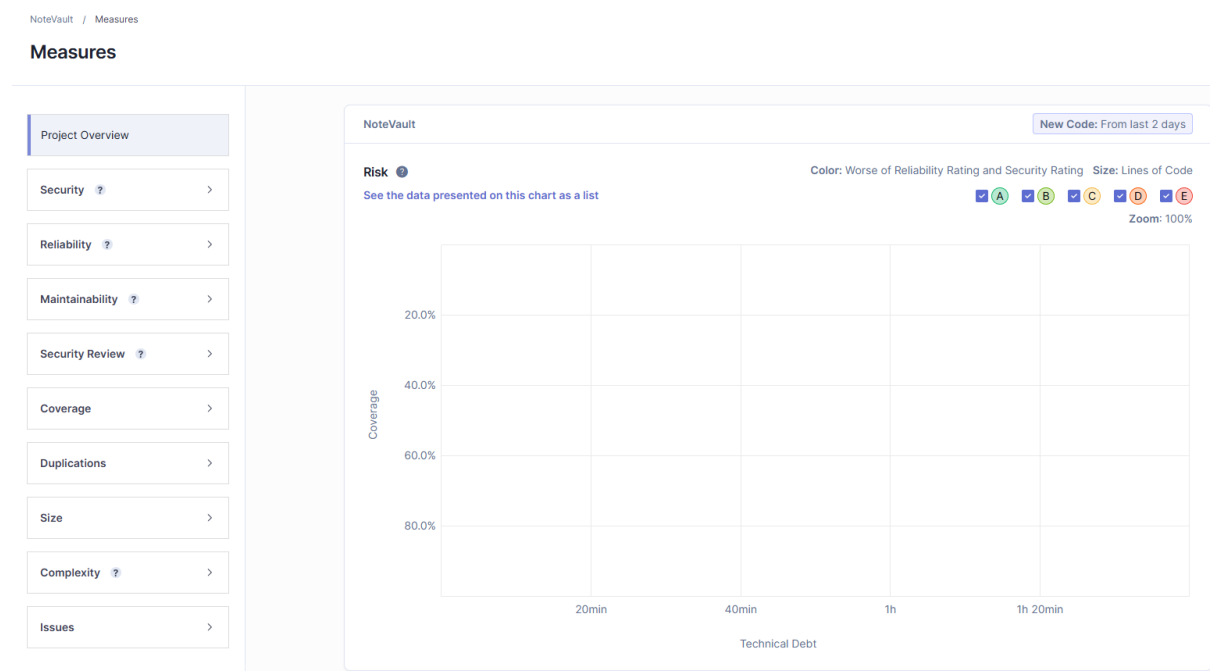


Figure 4. Screenshot of Security Hotspots

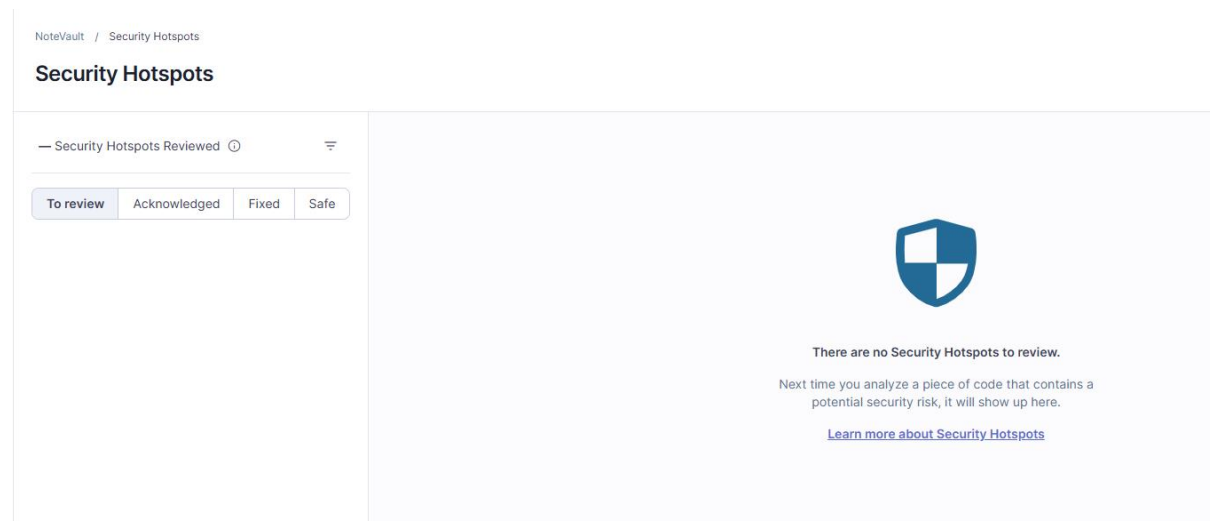
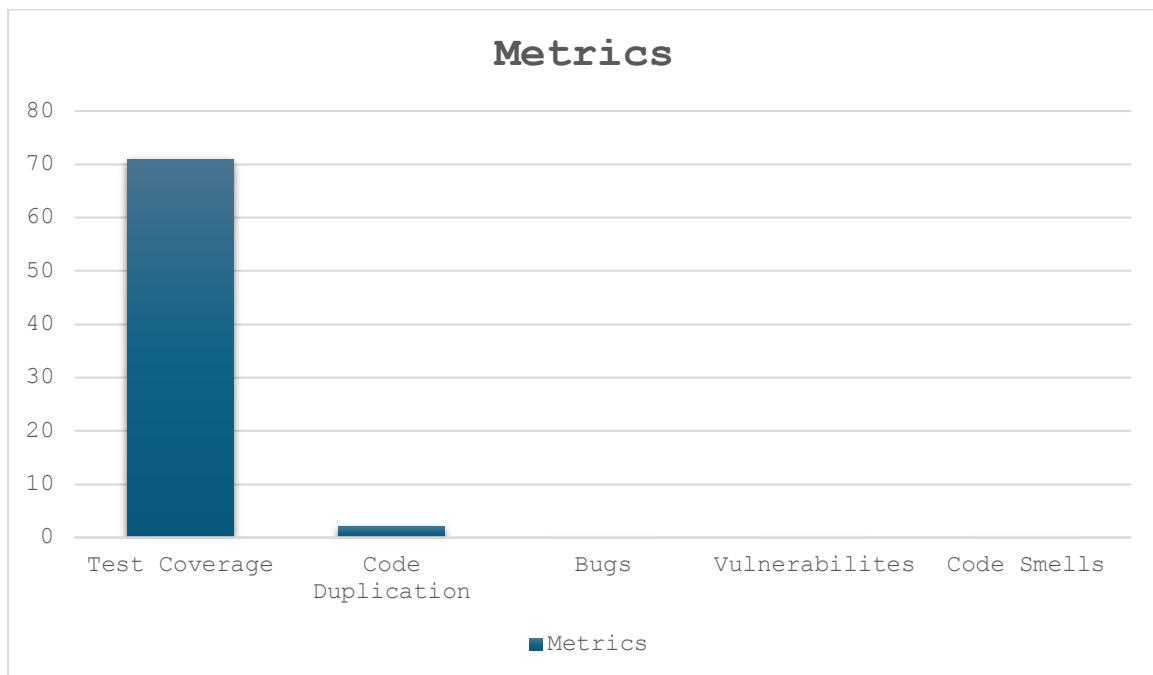


Figure 5. Software Quality Metrics Overview



9. Conclusion

The static code analysis performed using SonarQube provided a clear overview of the project's code quality, maintainability, and test coverage.

Overall, the project demonstrates strong code quality, with no detected bugs, vulnerabilities, or code smells. The security, reliability, and maintainability ratings are all A, indicating a well-structured and stable codebase.

However, the main weakness identified is low test coverage (71.7%), which is below the recommended threshold of 80%. This suggests that a significant portion of the code is not covered by automated tests, increasing the risk of undetected issues in untested areas.

To improve the project, increasing unit test coverage should be the highest priority, along with ensuring continuous testing during development.

In conclusion, while the codebase is of high quality in terms of structure and stability, improving test coverage would significantly enhance overall software reliability.